1. C PROGRAMMING BASICS TO WRITE A C PROGRAM:

Below are few commands and syntax used in C programming to write a simple C program. Let's see all the sections of a simple C program line by line.

C Basic commands	Explanation
#include <stdio.h>	This is a preprocessor command that includes standard input output header file(stdio.h) from the C library before compiling a C program
int main()	This is the main function from where execution of any C program begins.
{	This indicates the beginning of the main function.
/*_some_comments_*/	Whatever is given inside the command "/* */" in any C program, won't be considered for compilation and execution.
printf("Hello_World! ");	printf command prints the output onto the screen.
getch();	This command waits for any character input from keyboard.
return 0;	This command terminates C program (main function) and returns 0.
}	This indicates the end of the main function.

2. A SIMPLE C PROGRAM:

Below C program is a very simple and basic program in C programming language. This C program displays "Hello World!" in the output window. And, all syntax and commands in C programming are case sensitive. Also, each statement should be ended with semicolon (;) which is a statement terminator.

```
1 #include <stdio.h>
2 int main()
3 {
4    /* Our first simple C basic program */
5    printf("Hello World! ");
6    getch();
7    return 0;
8 }
```

OUTPUT:

```
Hello World!
```

3. STEPS TO WRITE C PROGRAMS AND GET THE OUTPUT:

Below are the steps to be followed for any C program to create and get the output. This is common to all C program and there is no exception whether it is a very small C program or very large C program.

1. Create (vi filename.c)
2. Compile (gcc filename.c)
3. Execute or Run (gcc –o file filename.c)
4. Get the Output (./a.out) (./file)

4. CREATION, COMPILATION AND EXECUTION OF A C PROGRAM:

Prerequisite:

- If you want to create, compile and execute C programs by your own, you have to install C compiler in your machine. Then, you can start to execute your own C programs in your machine.
- Once C compiler is installed in your machine, you can create, compile and execute C programs as shown in below link.
- If you don't want to install C/C++ compilers in your machine, you can refer online compilers which will compile and execute C/C++ and many other programming languages online and display outputs on the screen. Please search for online C/C++ compilers in Google for more details.

KEY POINTS TO REMEMBER IN C PROGRAMMING BASICS:

1. C programming is a case sensitive programming language.
2. Each C programming statement is ended with semicolon (;) which are referred as statement terminator.
3. printf() command is used to print the output onto the screen.
4. C programs are compiled using C compilers and displays output when executed.

C – printf and scanf

- printf() and scanf() functions are inbuilt library functions in C programming language which are available in C library by default. These functions are declared and related macros are defined in "stdio.h" which is a header file in C language.
- We have to include "stdio.h" file as shown in below C program to make use of these printf() and scanf() library functions in C language.

1. PRINTF() FUNCTION IN C LANGUAGE:

- In C programming language, printf() function is used to print the "character, string, float, integer, octal and hexadecimal values" onto the output screen.
- We use printf() function with %d format specifier to display the value of an integer variable.

- Similarly %c is used to display character, %f for float variable, %s for string variable, %lf for double and %x for hexadecimal variable.
- To generate a newline,we use "\n" in C printf() statement.

Note:

- C language is case sensitive. For example, printf() and scanf() are different from Printf() and Scanf(). All characters in printf() and scanf() functions must be in lower case.

EXAMPLE PROGRAM FOR C PRINTF () FUNCTION:

```
1  #include <stdio.h>
2  int main()
3  {
4     char ch = 'A';
5     char str[11] = "joysamadder";
6     float flt = 10.234;
7     int no = 150;
8     double dbl = 20.123456;
9     printf("Character is %c \n", ch);
10    printf("String is %s \n", str);
11    printf("Float value is %f \n", flt);
12    printf("Integer value is %d\n" , no);
13    printf("Double value is %lf \n", dbl);
14    printf("Octal value is %o \n", no);
15    printf("Hexadecimal value is %x \n", no);
16    return 0;
17 }
```

OUTPUT:

```
Character is A
String is joysamadder
Float value is 10.234000
Integer value is 150
Double value is 20.123456
Octal value is 226
Hexadecimal value is 96
```

You can see the output with the same data which are placed within the double quotes of printf statement in the program except

- %d got replaced by value of an integer variable (no),
- %c got replaced by value of a character variable (ch),
- %f got replaced by value of a float variable (flt),
- %lf got replaced by value of a double variable (dbl),
- %s got replaced by value of a string variable (str),
- %o got replaced by a octal value corresponding to integer variable (no),

- %x got replaced by a hexadecimal value corresponding to integer variable
- \n got replaced by a newline.

2. SCANF() FUNCTION IN C LANGUAGE:

- In C programming language, scanf() function is used to read character, string, numeric data from keyboard
- Consider below example program where user enters a character. This value is assigned to the variable "ch" and then displayed.
- Then, user enters a string and this value is assigned to the variable "str" and then displayed.

EXAMPLE PROGRAM FOR PRINTF() AND SCANF() FUNCTIONS IN C PROGRAMMING LANGUAGE:

```
1  #include <stdio.h>
2  int main()
3  {
4     char ch;
5     char str[100];
6     printf("Enter any character \n");
7     scanf("%c", &ch);
8     printf("Entered character is %c \n", ch);
9     printf("Enter any string ( upto 100 character ) \n");
10    scanf("%s", &str);
11    printf("Entered string is %s \n", str);
12 }
```

OUTPUT:

```
Enter any character
a
Entered character is a
Enter any string ( upto 100 character )
hai
Entered string is hai
```

- The format specifier %d is used in scanf() statement. So that, the value entered is received as an integer and %s for string.
- Ampersand is used before variable name "ch" in scanf() statement as &ch.
- It is just like in a pointer which is used to point to the variable.
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KEY POINTS TO REMEMBER IN C PRINTF() AND SCANF():

1. printf() is used to display the output and scanf() is used to read the inputs.
2. printf() and scanf() functions are declared in "stdio.h" header file in C library.
3. All syntax in C language including printf() and scanf() functions are case sensitive.